Review

Emulating Human-like Adaptive Vision for Efficient and Flexible Machine Visual Perception

Yulin Wang, Yang Yue et al.

Presentación DLfCV

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Objetivos y preguntas

1. Entender el concepto de Adaptive Neural Networks
2. Entender la arquitectura del modelo
3. Entender los resultados
4. Explorar el código

¿Las Adaptive NN modifican weights en runtime o usar distintos modelos ya lo hace una Adaptive NN?
¿Como es la arquitectura del modelo de baja resolución (glance)?
¿Qué datos usaron para el entrenamiento del modelo de baja resolución?
¿Cómo definen las regiones de interés?

1 . Revisión general

Lab: LEAP

Año: Septiembre 2025

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Problema detectado: Los algoritmos de visión más avanzados llegaron a un punto donde es imposible ejectuarlos en escenarios edge (robótica, wereables o escenas de riesgo)

Solución propuesta: ".. cambiar el pradigma de los modelos de visión de "pasivos" a "activos" "
Emular los procesos neurológicos de percepción visual.
Ir de una percepción general a focalizada de forma secuencial

Notas destacadas: Mencionan que su solución es "novel"
Mencionan que reducen el costo de inferencia de modelos por 28x sin sacrificar accuracy
Mencionan que el coportamiento de AdaptiveNN es indistinguible de humanos en algunos casos

2. Procesamiento visual en el cerebro

Sacadas - Fovea

https://vrx.vr-expert.com/wp-content/uploads/2022/08/ezgif.com-gif-maker-2-1568x684.webp

Fig 29-8 pg 646

Kandel, E. R., Koester, J., Mack, S., & Siegelbaum, S. (Eds.). (2021). Principles of neural science (Sixth edition). McGraw Hill.

2. Procesamiento visual en el cerebro

Integración de procesos

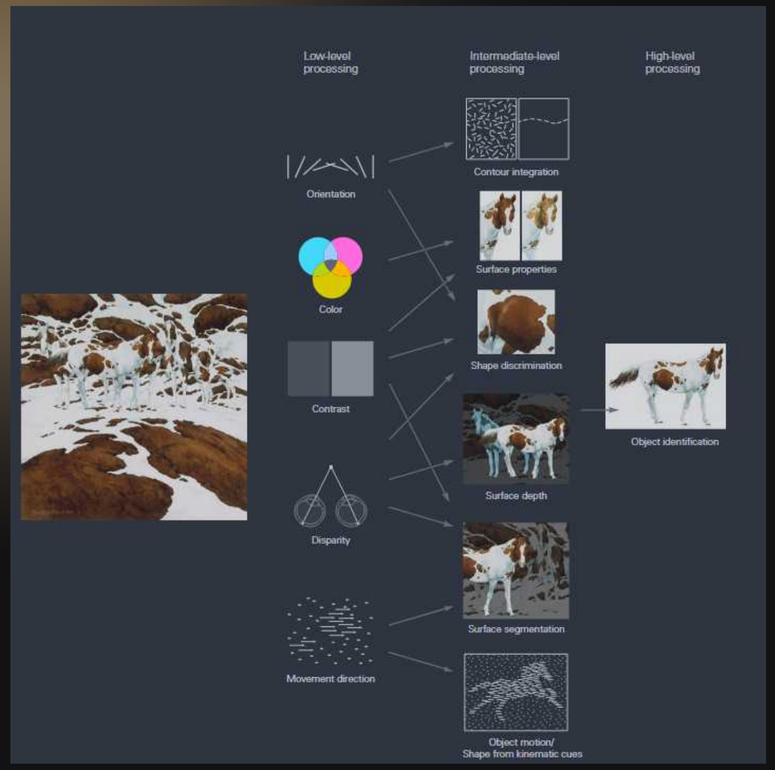

Fig 25-4 pg 560
Kandel, E. R., Koester, J., Mack, S., & Siegelbaum, S. (Eds.). (2021). Principles of neural science (Sixth edition). McGraw Hill.

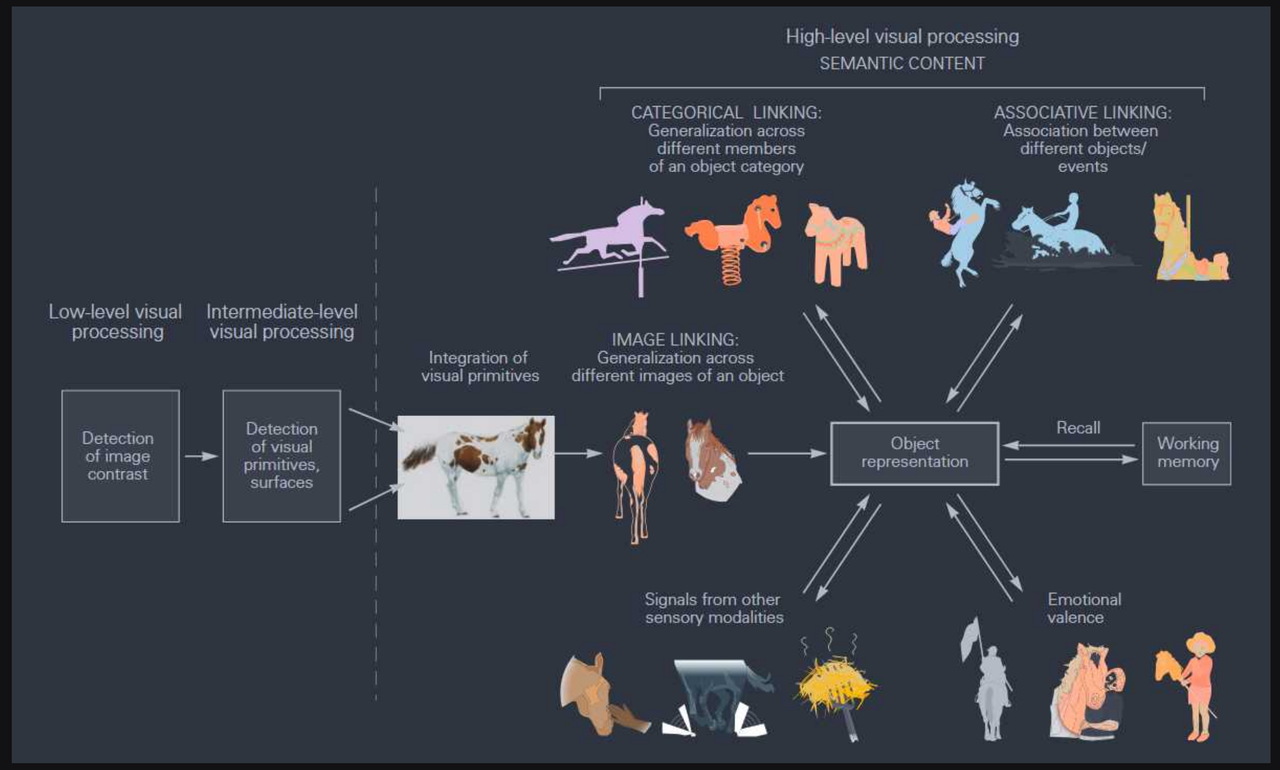

Fig 28-1 pg 622
Kandel, E. R., Koester, J., Mack, S., & Siegelbaum, S. (Eds.). (2021). Principles of neural science (Sixth edition). McGraw Hill.

3. Propuesta del paper

Presentan un análisis que **integra el aprendizaje de representaciones (feature learning) con el aprendizaje por refuerzo autorrecompensado**, lo que permite entrenar la el modelo sin depender de estructuras especializadas ni de anotaciones adicionales más allá de los objetivos estándar.

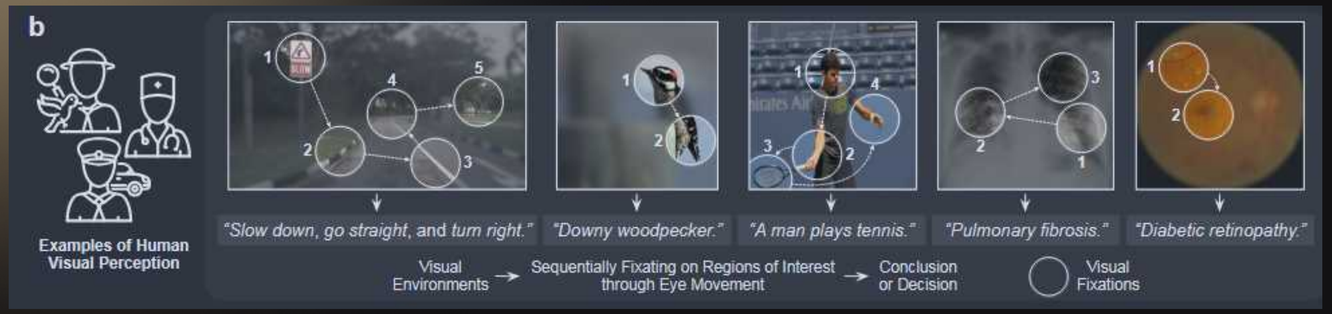

Fig 1. (b)
Wang, Y., Yue, Y., et al. (2025).
Emulating Human-like Adaptive Vision for Efficient and Flexible Machine Visual Perception (arXiv:2509.15333). arXiv. https://doi.org/10.48550/arXiv.2509.15333

Formula una secuencia de percepción mediante procesos dinámicos de toma de decisiones, fijandose unicamente en zonas de interés relevantes a la tarea a ejecutar.
Actualizando una representación interna que se refina hasta que el modelo decide que es suficiente información..

3. Arquitectura

- Un **modelo de base** (ResNet, Deit, OpenFlamingo) analiza una versión de menos resolución de la imagen para detectar features

- Un **modelo de reinforcement learning** entrenado para detectar los puntos de interés de la imagen genera una lista de puntos de interés.

- Un **agente de visión** que toma esos puntos de interéss y va seleccionando de esa lista (de a una o varias). Y hace inferencia de contenido de ese patch. Comprueba si la información que tiene es suficiente para resolver la tarea. Si no decide que no tiene suficiente información, toma otro patch.

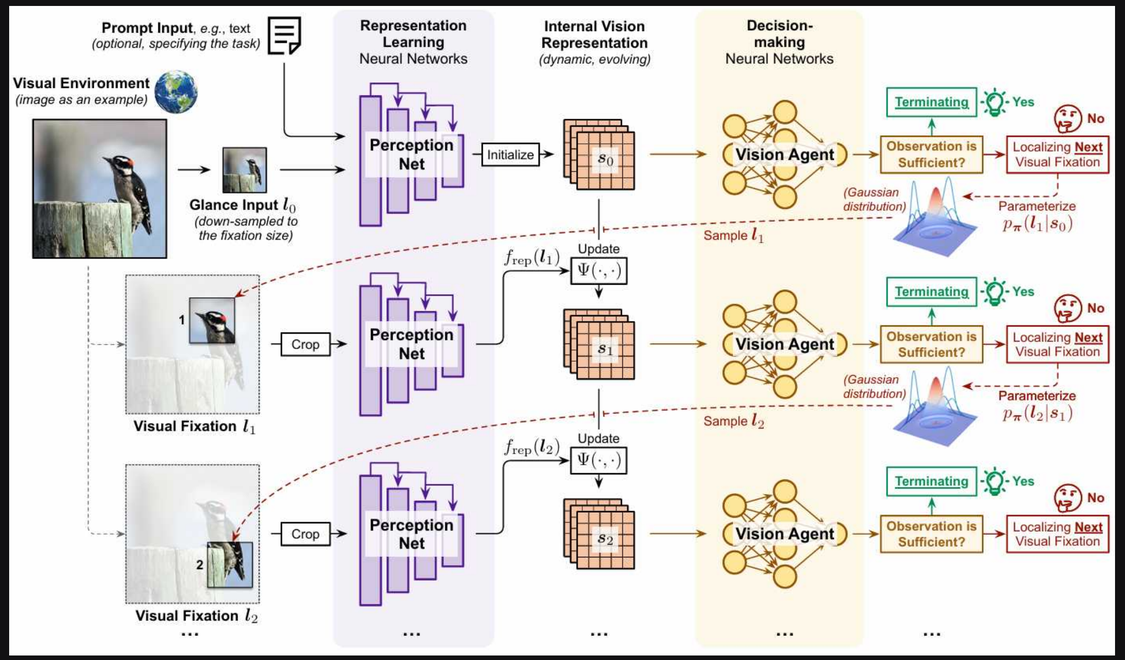

Fig 2. (a)
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Emulating Human-like Adaptive Vision for Efficient and Flexible Machine Visual Perception (arXiv:2509.15333). arXiv. https://doi.org/10.48550/arXiv.2509.15333

5. Preguntas

¿Las Adaptive NN modifican weights en runtime o usar distintos modelos ya lo hace una Adaptive NN?

- No. AdaptiveNN puede ajustar la cantidad de fijaciones que se utilizan previo a correr la inferencia

¿Como es la arquitectura del modelo de baja resolución?

- Se utilizan los primeros 3 bloques de ResNet o los primeros 8 bloques de DeiT

¿Qué datos usaron para el entrenamiento del modelo de baja resolución?

- El pipeline entero se entrenaba para un dataset específico.
  Se utilizaron: ImageNet-1K, CUB-200-2011, NABirds, Oxford-IIIT Pet, Stanford Dogs, Stanford Cars,FGVC-Aircraft, Swedish Traffic Signs Dataset, Synthetic Multi-Digit Search Dataset (MNIST), RSNA Pneumonia Detection Dataset, CALVIN Benchmark

¿Cómo definen las regiones de interés?

- Durante el entrenamiento el algoritmo de RL aprende a detectar zonas de mayor relevancia desde los features.

¿En que se diferencia con ViT?

- El hecho que ambos trabajen con parches puede confundir. La diferencia principal es que AdaptiveNN no depende del proceso de embedding de la imagen. Puede ser utilizado sobre esa arquitectura, pero no es excluyente.

4. Pruebas ejecutadas en paper

Costo computacional

Busqueda

Acción multimodal (robótica)

Comparación con humanos

4. Pruebas ejecutadas en paper

Costo computacional

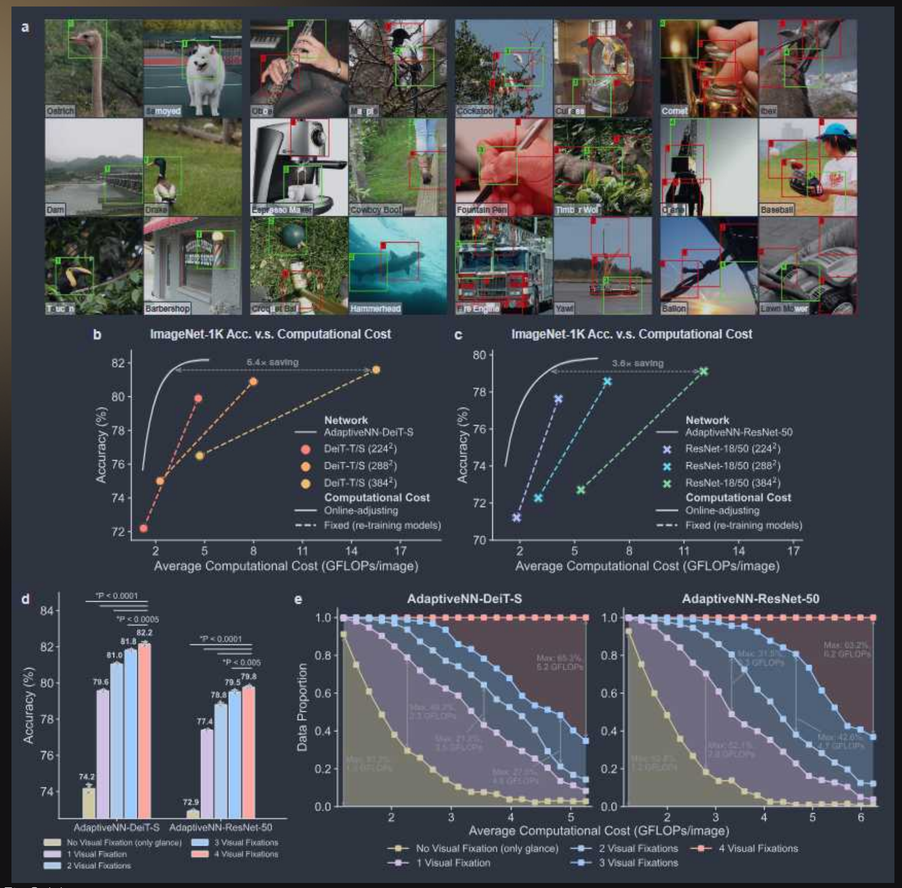

Fig 2. (a)
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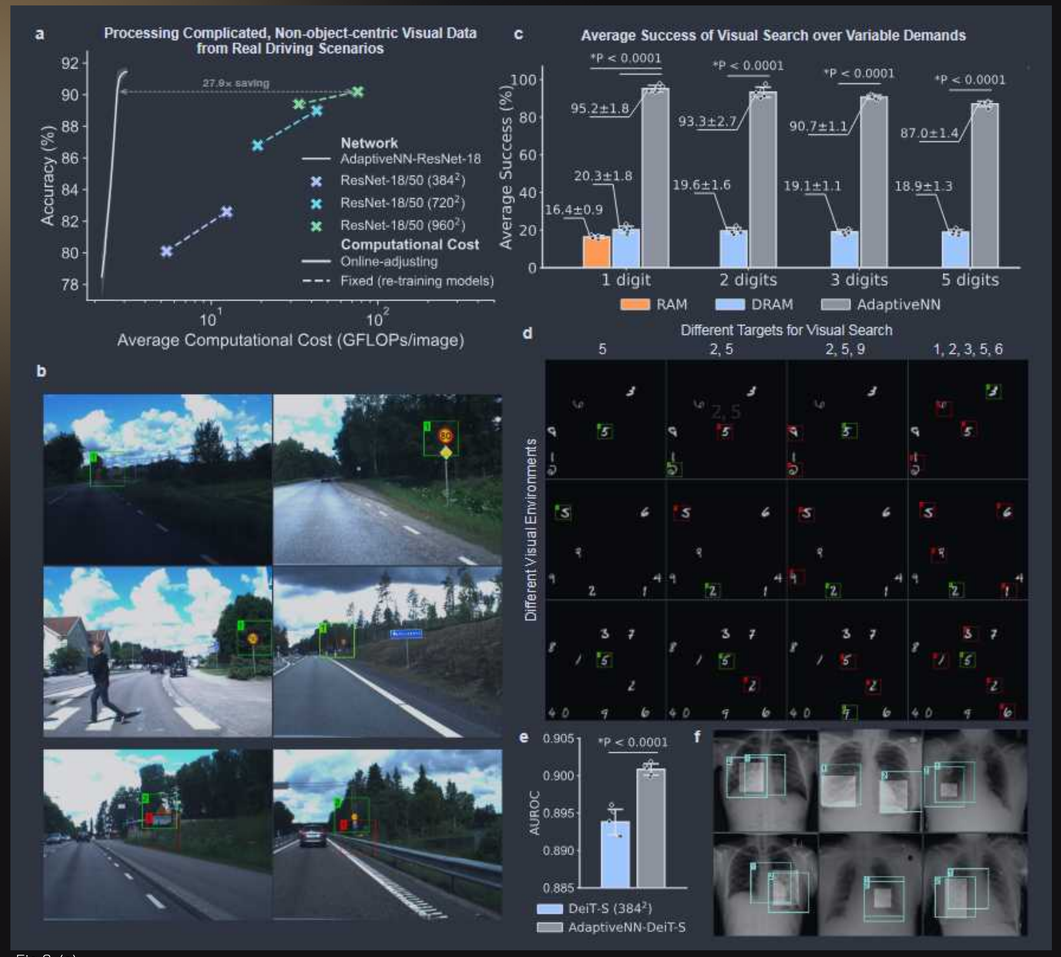
4. Pruebas ejecutadas en paper

Búsqueda



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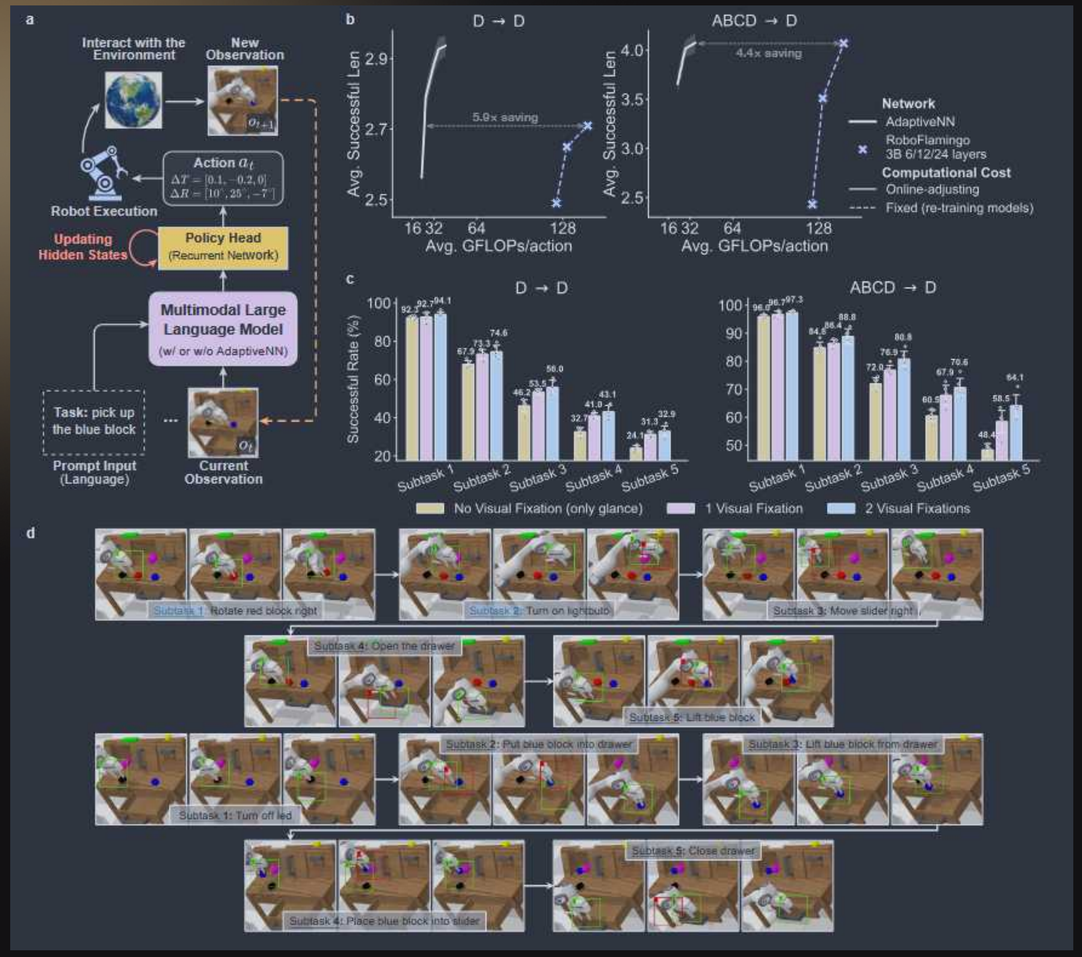
Model	Size of Image (fixation)	Computational Cost (GFLOPs/image)	Accuracy (%)
Models with fixed computational cost.			
ResNet-18	384^2 (–)	5.44	80.1
ResNet-18	720^2 (–)	18.98	86.8
ResNet-18	960^2 (–)	33.50	89.4
ResNet-50	384^2 (–)	12.38	82.6
ResNet-50	720^2 (–)	42.79	89.0
ResNet-50	960^2 (–)	75.88	90.2
AdaptiveNN with online-adjustable computational cost.			
AdaptiveNN-ResNet-18 (ours)	960 × 1280 (192^2)	2.20	78.5 ± 1.39
		2.29	80.0 ± 1.41
		2.38	82.1 ± 1.11
		2.47	83.9 ± 1.10
		2.56	86.0 ± 1.22
		2.65	88.4 ± 1.27
		2.74	90.4 ± 1.09
		2.83	91.1 ± 0.49
		2.92	91.3 ± 0.36
		3.01	91.4 ± 0.30
		3.10	91.5 ± 0.28

Supplementary Data Tab. 16 Comparisons of AdaptiveNN and traditional non-adaptive models in processing complicated, non-object-centric real-world scenes: Top-1 validation accuracy versus average computational cost for inferring the model. We consider the traffic sign recognition task on the Swedish traffic signs dataset (STSD), composed of 960x1,280 road-scene images collected on real moving vehicles.

4. Pruebas ejecutadas en paper

Acción multimodal (robótica)



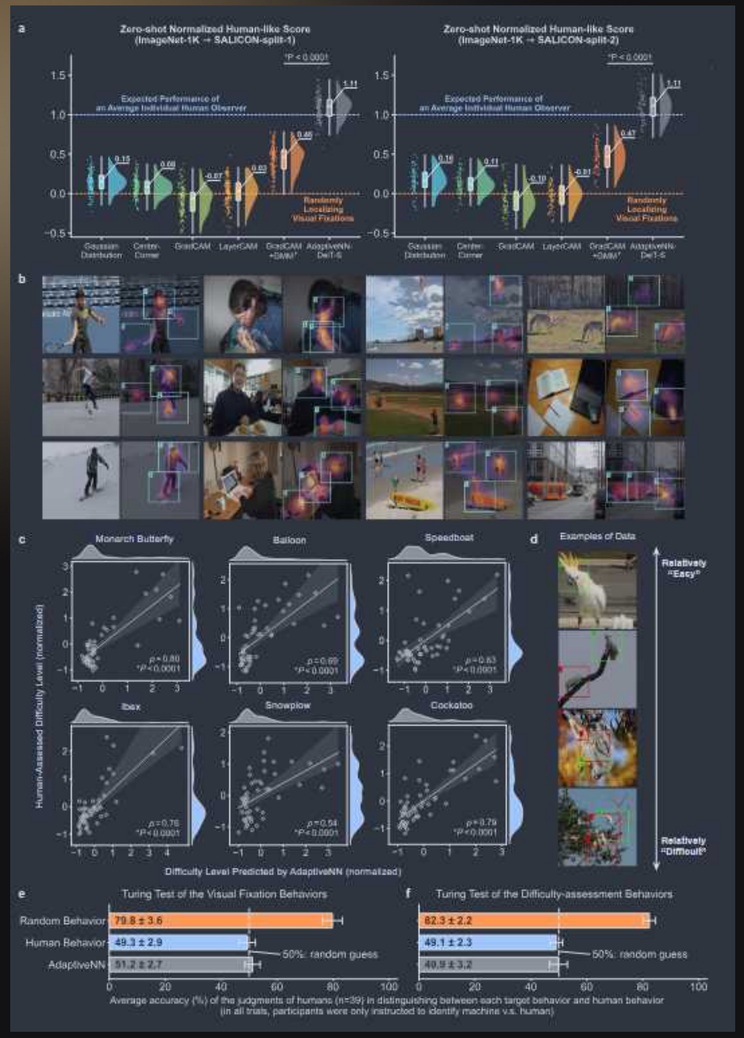
Model	Size of Image (fixation)	#Visual Fixations (average)	Computational Cost (GFLOPs/action)	Avg. Successful Length
*Models with **fixed** computational cost.*				
RoboFlamingo 6 layers	224^2 (–)	–	123.2	2.43
RoboFlamingo 12 layers	224^2 (–)	–	131.0	3.51
RoboFlamingo 24 layers	224^2 (–)	–	146.6	4.07
*AdaptiveNN with **online-adjustable** computational cost.*				
AdaptiveNN (ours)	224^2 (70^2)	0.13	22.8	3.66 ± 0.10
		0.20	23.7	3.71 ± 0.11
		0.28	24.7	3.78 ± 0.10
		0.38	25.9	3.84 ± 0.12
		0.43	26.5	3.90 ± 0.11
		0.61	28.7	4.02 ± 0.09
		0.78	30.8	4.03 ± 0.06
		0.95	32.9	4.05 ± 0.06
		1.20	35.9	4.07 ± 0.10

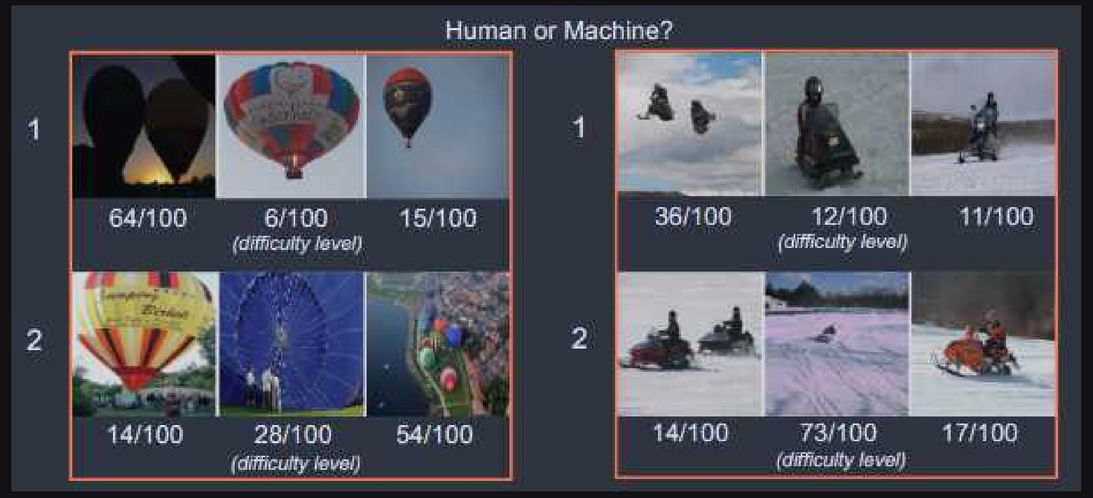
Supplementary Data Tab. 20 Quantitative comparisons of AdaptiveNN-based MLLM and non-adaptive MLLM on CALVIN: average successful length for ABCD→D versus average computational cost for inferring the model. For the non-adaptive models, computational costs are modulated by adjusting model sizes.

Fig 2. (a)
Wang, Y., Yue, Y., et al. (2025).
Emulating Human-like Adaptive Vision for Efficient and Flexible Machine Visual Perception (arXiv:2509.15333). arXiv. https://doi.org/10.48550/arXiv.2509.15333

4. Pruebas ejecutadas en paper

Comparación con humanos





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Emulating Human-like Adaptive Vision for Efficient and Flexible Machine Visual Perception (arXiv:2509.15333). arXiv. https://doi.org/10.48550/arXiv.2509.15333

5. Pruebas propias

Notas sobre repo

Descargar checkpoint
https://drive.google.com/file/d/1OIPZyTozJatdd40VZKaJHsw0ShaujQ0Y/view

File central
models\dynamic_deitS.py

5. Pruebas propias

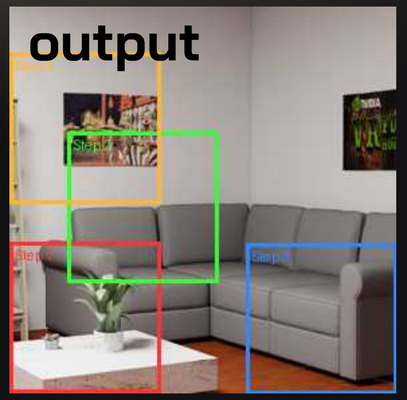
Notas sobre repo

- **Información de checkpoint base**
  Arquitectura base: Data-efficient Image Transformer - Small (Touvron et al., ICML 2021).
- **Pre entrenado en con** ImageNet-1K (ILSVRC2012)

Model Attribute (dynamic_deitS.py)	Tensors	Exact Parameters	Approx
focus_net	151	21,993,448	~22.0M
glance_net	151	21,993,448	~22.0M
offset_xy	1	50	~0.0M
policy_net_classifier_list	192	28,391,424	~28.4M
policy_net_head	10	1,925,000	~1.9M
policy_net_norm	10	3,840	~0.0M
policy_net_patch	21	7,012,736	~7.0M
policy_net_patch_norm_list	8	3,072	~0.0M
policy_net_patch_norm_list_stateV	8	3,072	~0.0M
policy_net_patch_stateV	21	7,012,224	~7.0M
policy_net_recover	4	629,785	~0.6M
policy_net_recover_preLN_list	8	3,072	~0.0M
policy_net_recover_reuseConvHead	1	18,816	~0.0M
policy_net_recover_reuseMLP	4	1,181,568	~1.2M
policy_net_recover_reuseMLP_preLN_list	8	3,072	~0.0M
policy_net_recover_reuseNorm_list	8	3,072	~0.0M
GRAND TOTAL	606	90,177,699	~90.2M

5. Pruebas propias

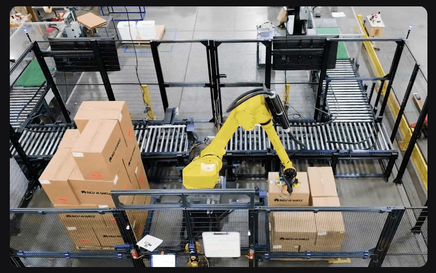
Resultados

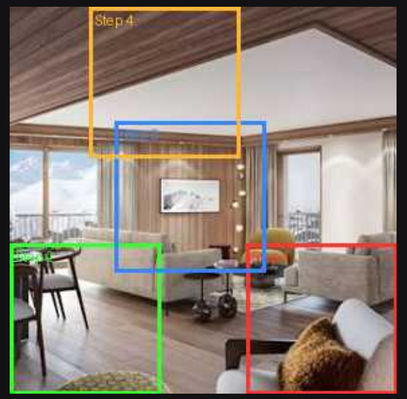


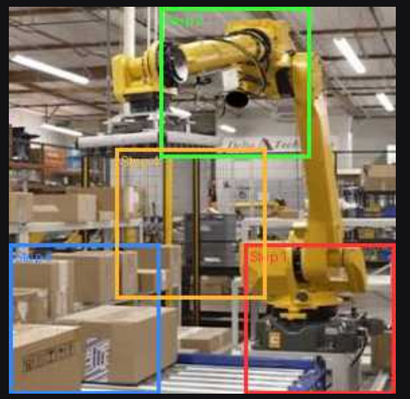

```
============================================================
STAGE 1: GLANCE INFERENCE (Low-res 112x112 Coarse Pass)
============================================================
  #1: studio couch                (ID: 831 ) | Confidence:  85.99%
  #2: window shade                (ID: 905 ) | Confidence:   0.81%
  #3: bookcase                    (ID: 453 ) | Confidence:   0.40%
  #4: table lamp                  (ID: 846 ) | Confidence:   0.29%
  #5: home theater                (ID: 598 ) | Confidence:   0.29%

============================================================
STAGE 2: SEQUENTIAL FOCUS REFINEMENT (4 Steps)
============================================================
  Step 1: Policy Crop Center (y=0.996, x=0.000)
          Box: [0, 175] -> [112, 287] (112x112 px)
          Top: studio couch | Confidence: 76.25%

  Step 2: Policy Crop Center (y=0.530, x=0.247)
          Box: [43, 93] -> [155, 205] (112x112 px)
          Top: studio couch | Confidence: 85.63%

  Step 3: Policy Crop Center (y=1.000, x=1.000)
          Box: [176, 176] -> [288, 288] (112x112 px)
          Top: studio couch | Confidence: 85.87%

  Step 4: Policy Crop Center (y=0.198, x=0.000)
          Box: [0, 34] -> [112, 146] (112x112 px)
          Top: studio couch | Confidence: 87.75%

============================================================
FINAL CLASSIFICATION (Aggregated Focus Output)
============================================================
  #1: studio couch                (ID: 831 ) | Confidence:  87.75%
  #2: bookcase                    (ID: 453 ) | Confidence:   0.55%
  #3: television                  (ID: 851 ) | Confidence:   0.45%
  #4: table lamp                  (ID: 846 ) | Confidence:   0.31%
  #5: home theater                (ID: 598 ) | Confidence:   0.30%
============================================================
```

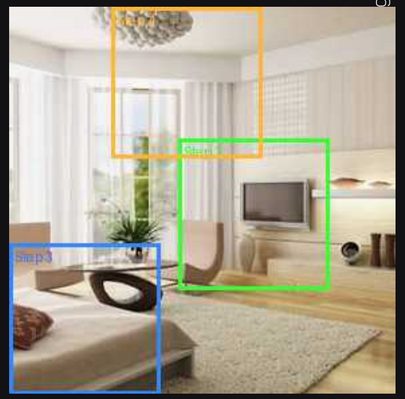
5. Pruebas propias

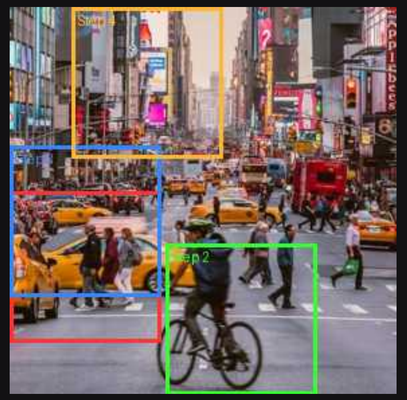
Resultados







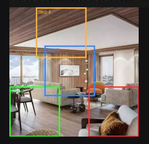




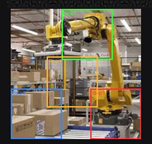
5. Pruebas propias

Resultados

```
============================================================
STAGE 1: GLANCE INFERENCE (Low-res 112x112 Coarse Pass)
============================================================
  #1: window shade                (ID: 905 ) | Confidence:  72.72%
  #2: sliding door                (ID: 799 ) | Confidence:   3.77%
  #3: studio couch                (ID: 831 ) | Confidence:   1.61%
  #4: shoji                       (ID: 789 ) | Confidence:   1.29%
  #5: window screen               (ID: 904 ) | Confidence:   1.05%

============================================================
STAGE 2: SEQUENTIAL FOCUS REFINEMENT (4 Steps)
============================================================
  Step 1: Policy Crop Center (y=1.000, x=1.000)
          Box: [176, 176] -> [288, 288] (112x112 px)
          Top: window shade | Confidence: 73.61%

  Step 2: Policy Crop Center (y=1.000, x=0.000)
          Box: [0, 176] -> [112, 288] (112x112 px)
          Top: window shade | Confidence: 73.05%

  Step 3: Policy Crop Center (y=0.483, x=0.447)
          Box: [78, 85] -> [190, 197] (112x112 px)
          Top: window shade | Confidence: 68.83%

  Step 4: Policy Crop Center (y=0.000, x=0.340)
          Box: [59, 0] -> [171, 112] (112x112 px)
          Top: window shade | Confidence: 67.86%

============================================================
FINAL CLASSIFICATION (Aggregated Focus Output)
============================================================
  #1: window shade                (ID: 905 ) | Confidence:  67.86%
  #2: studio couch                (ID: 831 ) | Confidence:   3.89%
  #3: sliding door                (ID: 799 ) | Confidence:   3.84%
  #4: shoji                       (ID: 789 ) | Confidence:   2.04%
  #5: restaurant                  (ID: 762 ) | Confidence:   1.88%
============================================================
```



```
============================================================
STAGE 1: GLANCE INFERENCE (Low-res 112x112 Coarse Pass)
============================================================
  #1: power drill                 (ID: 740 ) | Confidence:  45.56%
  #2: crane                       (ID: 517 ) | Confidence:  38.54%
  #3: forklift                    (ID: 561 ) | Confidence:   1.77%
  #4: lumbermill                  (ID: 634 ) | Confidence:   0.78%
  #5: chain saw                   (ID: 491 ) | Confidence:   0.68%

============================================================
STAGE 2: SEQUENTIAL FOCUS REFINEMENT (4 Steps)
============================================================
  Step 1: Policy Crop Center (y=1.000, x=0.997)
          Box: [175, 176] -> [287, 288] (112x112 px)
          Top: crane | Confidence: 40.02%

  Step 2: Policy Crop Center (y=0.000, x=0.637)
          Box: [112, 0] -> [224, 112] (112x112 px)
          Top: crane | Confidence: 45.03%

  Step 3: Policy Crop Center (y=1.000, x=0.000)
          Box: [0, 176] -> [112, 288] (112x112 px)
          Top: crane | Confidence: 33.48%

  Step 4: Policy Crop Center (y=0.598, x=0.454)
          Box: [79, 105] -> [191, 217] (112x112 px)
          Top: lumbermill | Confidence: 22.75%

============================================================
FINAL CLASSIFICATION (Aggregated Focus Output)
============================================================
  #1: lumbermill                  (ID: 634 ) | Confidence:  22.75%
  #2: crane                       (ID: 517 ) | Confidence:  19.03%
  #3: forklift                    (ID: 561 ) | Confidence:   6.85%
  #4: hook                        (ID: 600 ) | Confidence:   5.51%
  #5: guillotine                  (ID: 583 ) | Confidence:   3.81%
============================================================
```



```
============================================================
STAGE 1: GLANCE INFERENCE (Low-res 112x112 Coarse Pass)
============================================================
  #1: cab                         (ID: 468 ) | Confidence:  60.03%
  #2: unicycle                    (ID: 880 ) | Confidence:   6.34%
  #3: jinrikisha                  (ID: 612 ) | Confidence:   3.47%
  #4: moped                       (ID: 665 ) | Confidence:   2.85%
  #5: tricycle                    (ID: 870 ) | Confidence:   1.48%

============================================================
STAGE 2: SEQUENTIAL FOCUS REFINEMENT (4 Steps)
============================================================
  Step 1: Policy Crop Center (y=0.781, x=0.000)
          Box: [0, 137] -> [112, 249] (112x112 px)
          Top: cab | Confidence: 51.81%

  Step 2: Policy Crop Center (y=1.000, x=0.663)
          Box: [116, 176] -> [228, 288] (112x112 px)
          Top: cab | Confidence: 73.79%

  Step 3: Policy Crop Center (y=0.588, x=0.000)
          Box: [0, 103] -> [112, 215] (112x112 px)
          Top: cab | Confidence: 64.45%

  Step 4: Policy Crop Center (y=0.001, x=0.262)
          Box: [46, 0] -> [158, 112] (112x112 px)
          Top: cab | Confidence: 68.12%

============================================================
FINAL CLASSIFICATION (Aggregated Focus Output)
============================================================
  #1: cab                         (ID: 468 ) | Confidence:  68.12%
  #2: jinrikisha                  (ID: 612 ) | Confidence:   2.22%
  #3: moped                       (ID: 665 ) | Confidence:   0.45%
  #4: traffic light               (ID: 920 ) | Confidence:   0.41%
  #5: street sign                 (ID: 919 ) | Confidence:   0.39%
============================================================
```

6. Conclusiones y posibles proximos pasos

Conclusiones

- Según los resultados presentados en el paper, parece que esta técnica cumple con el objetivo de reducir costo de computo y se asemeja a procesos cognitivos atencionales
- Se me hizo difícil poder entender del todo el algoritmo, principalmente por uso de técnicas y funciones que no vimos en el curso.
- Corrí el modelo de prueba que ofrecen en una GPU 4070 y corre en dos segundos aprox.
- Los resultados están limitados por el dataset con el que entrenaron.
- LEAP Lab es buen referente

Posibles próximos pasos

- Basado en la semejanza de esta técnica con procesos cognitivos atencionales. Me intriga como se comportará en experimentos atencionales como la tarea de Posner
- Me intriga como se podrían usar modelos de saliencia visual como modelos base para esta técnica.

Gracias

Preguntas?